Bolt Tightening Inspection
for Piping and Structure Procedure

(Doc No. CT8-PRO-CM-008)

USA PROJECT UNIT1 - TALLGRASS

Revision History				
Revisions	*Date*	*Descriptions*	*Prepared By*	*Approved By*
0	01-Apr-26	First Issue	Ms. Michelle Wilrutstien	Mr. Phisit Pew-la-or

Prepared by (QA Engineer):	Ms. Michelle Wilrutstien	Approved by (QAQC Manager):	Mr. Phisit Pew-la-or	Approved by (Customer):	REVIEWED BY QC INSPECTOR SIGNATURE: DATE: CIMTAS NINGBO Digitally signed ~~by Suriya Lamor~~ Date: 2026.04.02 ~~11:04:24 +07'00'~~
Date:	01-Apr-26		01-Apr-26		
Signature:					

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1. SCOPE

This procedure defines the requirement for manual bolts tightening of steel structure and process piping assembly in accordance ASME B31.1 and MPW-A specifications.

2. REFERENCE

- ASME PCC-1, 2013: Guidelines for pressure boundary bolted flange joint assembly.
- ASME B31.1 Power Piping
- MSR-1005-011 STANDARDS AND RECOMMENDATIONS FOR TORQUE VALUES OF PIPE FLANGE GASKETS

3. RESPONSIBILITY

3.1 The BJCHI shall be responsible for performing the manual bolts tightening operation, in accordance with this procedure.

3.2 Piping supervisor shall be responsible for verifying torque value and stud bolt are correctly and tightened. flanges shall clearly identify / securely tag with matching. I.D. number to inspection check sheet. And Structure supervisor shall be checking structure bolt, type, sizing and grade comply with drawing before installation.

4. CALIBRATION

Manual torque wrench, hydraulic tightening gear and bolt tensioner shall have a calibration certificate valid for the date of task. In-situ calibration (on site) may require upon requested randomly by company representative. In case of in-situ calibration, bolt elongation shall be measured by bolt scope, UT narrow beam, caliper vernier or any methods which accepted by company, all equipment and tool for tightening to be calibration every 6 months.

5. BOLTS TORQUING FOR PIPING FLANGES

Unless otherwise specified, torque values shown in the following tables of this section may be used as a target value at the final installation of flanged connections after pressure test and cleaning.

It is important to note these charts are consistent with gasket crush values and are to be followed unless approval has otherwise been granted by MPW-A**.**

Required torque values will vary depending on multiple factors. The torque values presented therein do not necessarily guarantee a perfect installation of a flanged connection. Target torques may need to be adjusted depending on the property of actual thread lubricant used.

Torque values are based on the following conditions:

1. Use of low-friction thread lubricant or anti-seize, such as Lub-O-Seal NM91, on both fastener threads and head/nut bearing surface.

2. The thread friction coefficient and the under-head friction coefficient of 0.12 has been used for Tables 1-6, 8 and 9 and 0.13 has been used for Table 7, as a result of the use of thread lubricant or anti-seize.

3. Bolt axial stress of 35% of tensile strength for ASME flange except for Table 7 where 42% is used per API 6A, and 65% of tensile strength for SAE flange.

4. Tensile strength used for calculation is shown below:

Material	Tensile Strength
A193 Grade B7	860 N/mm^2

5. UNC thread for inch bolts up to 1" size and 8UNC thread for bolts larger than 1" size.

6. Standard coarse thread for metric bolts except for 3mm pitch for M30

Table 1: Torque Values for Class 150 Spiral Wound Gaskets

						Torque Value (Metric)		Torque Value (US)	
Class:150 Type:Spiral Wound Gasket	Nominal Flange Size (Metric)	Nominal Flange Size (US)	Quantity of Bolts	Bolt Size (Metric)	Bolt Size (US)	Liquid	Gas	Liquid	Gas
	(mm)	*(in)*		*(mm)*	*(in)*	(N-m)	*(N-m)*	*(Ft-lb)*	*(Ft-lb)*
	15	1/2	4	M12	1/2	11	25	8	18
	20	3/4	4	M12	1/2	16	36	12	27
	25	1	4	M12	1/2	22	50	16	37
	32	1 1/4	4	M12	1/2	23	53	17	39
	40	1 1/2	4	M12	1/2	33	76	24	56
	50	2	4	M16	5/8	54	122	40	90
	65	2 1/2	4	M16	5/8	62	142	46	105
	80	3	4	M16	5/8	91	207	67	153
	90	3 1/2	8	M16	5/8	51	115	38	85
	100	4	8	M16	5/8	67	152	49	112
	125	5	8	M20	3/4	95	217	70	160
	150	6	8	M20	3/4	136	312	100	230
	200	8	8	M20	3/4	193	442	142	326
	250	10	12	M22	7/8	182	416	134	307
	300	12	12	M22	7/8	248	567	183	418
	350	14	12	M24	1	309	706	228	521
	400	16	16	M24	1	312	714	230	527
	450	18	16	M27	1 1/8	504	1154	372	851
	500	20	20	M27	1 1/8	444	1017	327	750
	600	24	20	M30	1 1/4	644	1473	475	1086

Table 2: Torque Values for Class 300 Spiral Wound Gaskets

						Torque Value (Metric)		Torque Value (US)	
Class:300 Type:Spiral Wound Gasket	Nominal Flange Size (Metric)	Nominal Flange Size (US)	Quantity of Bolts	Bolt Size (Metric)	Bolt Size (US)	Liquid	Gas	Liquid	Gas
	(mm)	*(in)*		*(mm)*	*(in)*	*(N-m)*	*(N-m)*	*(Ft-lb)*	*(Ft-lb)*
	15	1/2	4	M12	1/2	11	25	8	18
	20	3/4	4	M16	5/8	20	45	15	33
	25	1	4	M16	5/8	28	62	21	46
	32	1 1/4	4	M16	5/8	36	87	27	64
	40	1 1/2	4	M20	3/4	59	114	44	84
	50	2	8	M16	5/8	34	61	25	45
	65	2 1/2	8	M20	3/4	52	85	38	63
	80	3	8	M20	3/4	76	124	56	91
	90	3 1/2	8	M20	3/4	80	138	59	102
	100	4	8	M20	3/4	115	182	85	134
	125	5	8	M20	3/4	153	217	113	160
	150	6	12	M20	3/4	142	268	105	198
	200	8	12	M22	7/8	246	341	181	252
	250	10	16	M24	1	290	357	214	263
	300	12	16	M27	1 1/8	443	548	327	404
	350	14	20	M27	1 1/8	409	477	302	352
	400	16	20	M30	1 1/4	585	715	431	527
	450	18	24	M30	1 1/4	628	856	463	631
	500	20	24	M30	1 1/4	739	942	545	695
	600	24	24	M36	1 1/2	1214	1471	895	1085
	650	26	28	M42	1 3/4	1830	1830	1350	1350
	800	32	28	M48	2	3254	3254	2400	2400

Table 3: Torque Values for Class 600 Spiral Wound Gaskets

						Torque Value (Metric)		Torque Value (US)	
Class:600 Type:Spiral Wound Gasket	Nominal Flange Size (Metric)	Nominal Flange Size (US)	Quantity of Bolts	Bolt Size (Metric)	Bolt Size (US)	Liquid	Gas	Liquid	Gas
	(mm)	*(in)*		*(mm)*	*(in)*	*(N-m)*	*(N-m)*	*(Ft-lb)*	*(Ft-lb)*
	15	1/2	4	M12	1/2	21	25	15	18
	20	3/4	4	M16	5/8	39	45	29	33
	25	1	4	M16	5/8	55	62	41	46
	32	1 1/4	4	M16	5/8	71	71	52	52
	40	1 1/2	4	M20	3/4	117	117	86	86
	50	2	8	M16	5/8	68	68	50	50
	65	2 1/2	8	M20	3/4	103	103	76	76
	80	3	8	M20	3/4	152	152	112	112
	90	3 1/2	8	M22	7/8	208	208	153	153
	100	4	8	M22	7/8	286	286	211	211
	125	5	8	M24	1	437	437	322	322
	150	6	12	M24	1	392	392	289	289
	200	8	12	M27	1 1/8	653	653	482	482
	250	10	16	M30	1 1/4	759	759	560	560
	300	12	20	M30	1 1/4	817	817	603	603
	350	14	20	M33	1 3/8	1022	1022	754	754
	400	16	20	M36	1 1/2	1428	1428	1053	1053
	450	18	20	M39	1 5/8	1965	1965	1449	1449
	500	20	24	M39	1 5/8	1925	1925	1420	1420
	600	24	24	M45	1 7/8	3012	3012	2222	2222

Table 4: Torque Values for Class 900 Spiral Wound Gaskets

						Torque Value (Metric)		Torque Value (US)	
Class:900 Type:Spiral Wound Gasket	Nominal Flange Size (Metric)	Nominal Flange Size (US)	Quantity of Bolts	Bolt Size (Metric)	Bolt Size (US)	Liquid	Gas	Liquid	Gas
	(mm)	*(in)*		*(mm)*	*(in)*	*(N-m)*	*(N-m)*	*(Ft-lb)*	*(Ft-lb)*
	15	1/2	4	M20	3/4	47	47	35	35
	20	3/4	4	M20	3/4	70	70	52	52
	25	1	4	M22	7/8	114	114	84	84
	32	1 1/4	4	M22	7/8	185	185	136	136
	40	1 1/2	4	M24	1	272	272	201	201
	50	2	8	M22	7/8	177	177	131	131
	65	2 1/2	8	M24	1	255	255	188	188
	80	3	8	M22	7/8	297	297	219	219
	100	4	8	M27	1 1/8	553	553	408	408
	125	5	8	M30	1 1/4	821	821	606	606
	150	6	12	M27	1 1/8	661	661	488	488
	200	8	12	M33	1 3/8	1134	1134	836	836
	250	10	16	M33	1 3/8	1171	1171	864	864
	300	12	20	M33	1 3/8	1295	1295	955	955
	350	14	20	M36	1 1/2	1628	1628	1201	1201
	400	16	20	M39	1 5/8	2227	2227	1643	1643
	450	18	20	M45	1 7/8	3326	3326	2453	2453
	500	20	20	M48	2	4125	4125	3042	3042
	600	24	20	M64	2 1/2	7023	7023	5180	5180

Table 5: Torque Values for Class 1500 Spiral Wound Gaskets

						Torque Value (Metric)		Torque Value (US)	
Class:1500 Type:Spiral Wound Gasket	Nominal Flange Size (Metric)	Nominal Flange Size (US)	Quantity of Bolts	Bolt Size (Metric)	Bolt Size (US)	Liquid	Gas	Liquid	Gas
	(mm)	*(in)*		*(mm)*	*(in)*	*(N-m)*	*(N-m)*	*(Ft-lb)*	*(Ft-lb)*
	15	1/2	4	M20	3/4	79	79	58	58
	20	3/4	4	M20	3/4	116	116	86	86
	25	1	4	M22	7/8	190	190	140	140
	32	1 1/4	4	M22	7/8	308	308	227	227
	40	1 1/2	4	M24	1	452	452	333	333
	50	2	8	M22	7/8	294	294	217	217
	65	2 1/2	8	M24	1	424	424	313	313
	80	3	8	M27	1 1/8	652	652	481	481
	100	4	8	M30	1 1/4	1042	1042	769	769
	125	5	8	M36	1 1/2	1680	1680	1239	1239
	150	6	12	M33	1 3/8	1363	1363	1005	1005
	200	8	12	M39	1 5/8	2293	2293	1691	1691
	250	10	12	M45	1 7/8	3675	3675	2711	2711
	300	12	16	M48	2	3916	3916	2888	2888
	350	14	16	M56	2 1/4	4977	4977	3671	3671
	400	16	16	M64	2 1/2	7240	7240	5340	5340
	450	18	16	M72	2 3/4	10150	10150	7486	7486
	500	20	16	M76	3	13033	13033	9613	9613
	600	24	16	M90	3 1/2	20915	20915	15426	15426

Table 6: Torque Values for Class 3000 SAE Hydraulic Flange O-ring

					Torque Value (Metric)		Torque Value (US)	
Class: 3000 Type: SAE Hydraulic Flange O-ring	Nominal Flange Size (Metric)	Nominal Flange Size (US)	Quantity of Cap Screws	Cap Screw Size (US)	Minimum Torque	Maximum Torque	Minimum Torque	Maximum Torque
	(mm)	*(in)*		*(in)*	*(N-m)*	*(N-m)*	*(Ft-lb)*	*(Ft-lb)*
	20	3/4	4	5/16	19.8	25.4	14.6	18.75
	25	1	4	3/8	36.7	48.0	27.1	35.4
	32	1 1/4	4	7/16	48.0	62.1	35.4	45.8
	40	1 1/2	4	1/2	107.4	124.3	79.2	91.7
	25	1	4	M10	36.7	48.0	27.1	35.4

Table 7: Torque Values for Class 600 Ring Joint Gaskets

					Torque Value (Metric)		Torque Value (US)	
Class: 600 Type: Ring Joint	Nominal Flange Size (Metric)	Nominal Flange Size (US)	Quantity of Bolts	Bolt Size (US)	Minimum Torque	Maximum Torque	Minimum Torque	Maximum Torque
	(mm)	*(in)*		*(in)*	*(N-m)*	*(N-m)*	*(Ft-lb)*	*(Ft-lb)*
	32	1 1/4	4	5/8	122.0	162.7	90	120
	25	1	4	5/8	122.0	162.7	90	120

Table 8: Torque Values for Class 150 Flat Ring, Compressed Non-Asbestos Fiber Gaskets

						Torque Value (Metric)		Torque Value (US)	
Class: 150 Type: Flat Ring, Compressed Non-Asbestos Fiber	Nominal Flange Size (Metric)	Nominal Flange Size (US)	Quantity of Bolts	Bolt Size (Metric)	Bolt Size (US)	Liquid	Gas	Liquid	Gas
	(mm)	*(in)*		*(mm)*	*(in)*	*(N-m)*	*(N-m)*	*(Ft-lb)*	*(Ft-lb)*
	15	1/2	4	M12	1/2	5	18	4	13
	20	3/4	4	M12	1/2	9	26	7	19
	25	1	4	M12	1/2	12	35	9	26
	32	1 1/4	4	M12	1/2	17	51	13	38
	40	1 1/2	4	M12	1/2	22	72	16	53
	50	2	4	M16	5/8	44	146	32	108
	65	2 1/2	4	M16	5/8	46	155	34	114
	80	3	4	M16	5/8	74	246	55	181
	90	3 1/2	8	M16	5/8	41	136	30	100
	100	4	8	M16	5/8	52	172	38	127
	125	5	8	M20	3/4	77	258	57	190
	150	6	8	M20	3/4	99	332	73	245
	200	8	8	M20	3/4	135	449	100	331
	250	10	12	M22	7/8	130	416	96	307
	300	12	12	M22	7/8	176	553	130	408
	350	14	12	M24	1	232	668	171	493
	400	16	16	M24	1	223	664	164	490
	450	18	16	M27	1 1/8	321	1015	237	749
	500	20	20	M27	1 1/8	303	893	223	659
	600	24	20	M30	1 1/4	463	1279	341	943

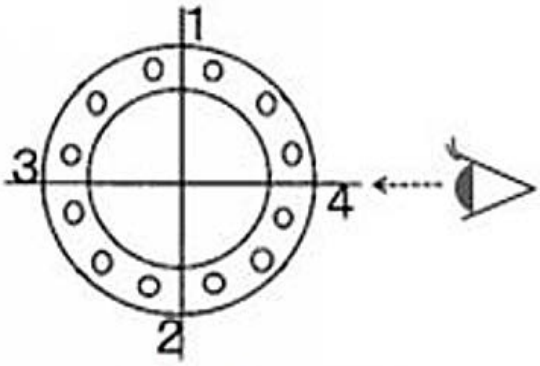
Table 9: Torque Values for Class 300 Flat Ring, Compressed Non-Asbestos Fiber Gaskets

						Torque Value (Metric)		Torque Value (US)	
Class:300 Type: Flat Ring, Compressed Non-Asbestos Fiber	Nominal Flange Size (Metric)	Nominal Flange Size (US)	Quantity of Bolts	Bolt Size (Metric)	Bolt Size (US)	Liquid	Gas	Liquid	Gas
	(mm)	*(in)*		*(mm)*	*(in)*	*(N-m)*	*(N-m)*	*(Ft-lb)*	*(Ft-lb)*
	15	1/2	4	M12	1/2	10	18	7	13
	20	3/4	4	M16	5/8	19	32	14	24
	25	1	4	M16	5/8	25	44	18	32
	32	1 1/4	4	M16	5/8	38	64	28	47
	40	1 1/2	4	M20	3/4	59	108	44	80
	50	2	8	M16	5/8	38	73	28	54
	65	2 1/2	8	M20	3/4	55	93	41	69
	80	3	8	M20	3/4	80	147	59	108
	90	3 1/2	8	M20	3/4	94	114	69	84
	100	4	8	M20	3/4	117	207	86	153
	125	5	8	M20	3/4	156	258	115	190
	150	6	12	M20	3/4	136	221	100	163
	200	8	12	M22	7/8	236	348	174	257
	250	10	16	M24	1	278	357	205	263
	300	12	16	M27	1 1/8	425	535	313	395
	350	14	20	M27	1 1/8	392	465	289	343
	400	16	20	M30	1 1/4	558	665	412	490
	450	18	24	M30	1 1/4	595	753	439	555
	500	20	24	M30	1 1/4	703	828	519	611
	600	24	24	M36	1 1/2	1158	1277	854	942

6. ALTERNATIVE METHOD OF GASKET INSTALLATION TORQUE VALUES

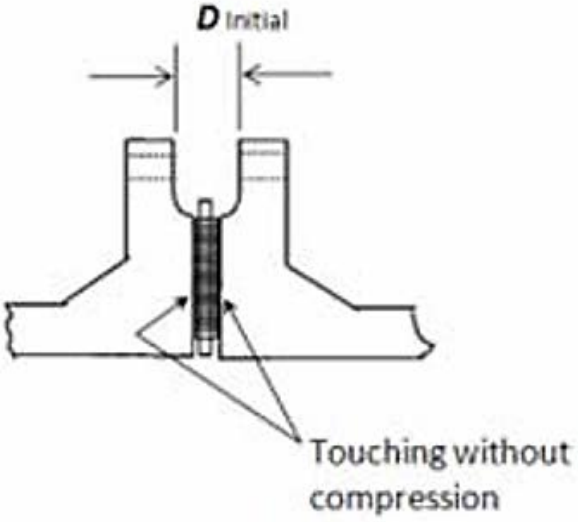
In the situation where a torque value is not provided, the process below can be used to determine final installation of a flanged connection with a spiral wound gasket.

1. During the rounds of bolt tightening, stop tightening when both sides of the gasket are almost in contact with the flange face. It should not be tight enough to compress the gasket or remove all lateral play from the gasket.
2. Check the gap between the gasket and the flange faces from four directions: top, bottom, left and right. When looking at the gap, you should not be able to see light through the flange faces.

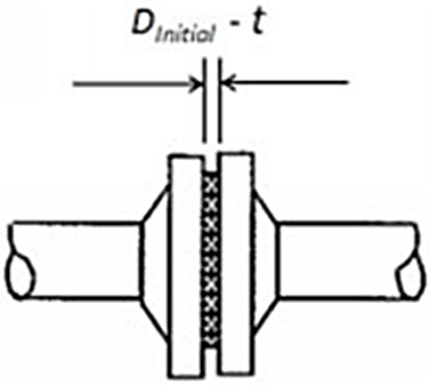


Check at four points

3. Measure the distance between the flanges at the four points. Record the measurements as **D**Initial.



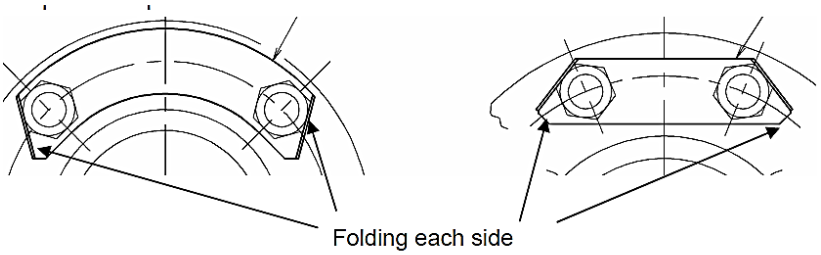
4. Start tightening the nuts in the sequence.
5. Continue the tightening until the total flange distance gets compressed by t = 0.8 mm to 1.2 mm (measured at the same four points).



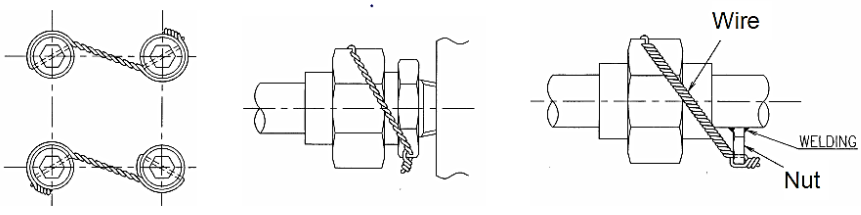
for spiral wound gasket

t = 0.8 to 1.2 mm

6. Final flange distance value **D** (**D**Initial – t) shall be verified at all four points to ensure a full uniform gasket seal.



<Lock wire>



(a) For square flange (b) For union (c) For union

7. After finishing all the test, the bidder shall be fixed all bolts **(**Anti-Rotation**)** is as above**.**

7. BOLTS TIGHTENING FOR STRUCTURE

Construction drawings shall indicate the grade and diameter of all bolts, nuts and washers required for the construction on the drawings. And all bolts tightening for structure shall be perform to snug type.

8. ATTACHMENTS

Attachment 1. Bolt Tightening Inspection Report-Piping

BJCHI

BOLT TIGHTENING INSPECTION REPORT - PIPING

cimtas | ningbo

PAGE 1 OF 1

Project Name :		BJCHI Job No.:	Inspection Date :
Client :		Procedure No.:	
Fabricator :	BJC Heavy Industries Public Company Limited	Module No.:	
Report No. :		Unit No. :	

FLANGE		STUD BOLT & GASKET DETAILS											
Spool No.	Joint No.	Bolt			Nut Grade	Gasket		TORQUE VALUE lbf-ft / in2				Torque Required	
		Grade	Dia.Bolt (inch)	Length (mm)		Type	Rating	1st PASS 30%	2nd PASS 60%	3rd PASS 100%	BOLTING RESIDUAL STRESS lbf-ft / in2	Ft-Lb	N.M

No.	ACTIVITY	INSPECTION RESULTS		
		Yes	No	Remark
1	Make sure flange faces clean & free from damage	☐	☐	
2	Flange face alignment complete & accepted	☐	☐	
3	Gasket type and condition verified and acceptable	☐	☐	
4	Thread protrusion pass the nut acceptable	☐	☐	
5	Bolt ends colour coded after completion of Tightening	☐	☐	

REMARKS:

RELEASED BY	Prepared QC BJCHI	QC Inspector BJCHI	CNBO	MPA
NAME :				
SIGNATURE :				
DATE :				

F-QA-220 Rev.1 (09 Jul 2025)_Bolt Tightening Inspection Report-Piping

Attachment 2. Bolt Connection & Gap Measurement

BJCHI **BOLT CONNECTION & GAP MEASUREMENT** Client Logo

Project Name :	
BJCHI Job No. :	
Client :	
Unit :	
Module :	
Inspection Date :	

Remarks:

Item No.	Description	Gap Measurement (mm)	Photo	Remarks

Result : ☐ Accepted ☐ Rejected

Witnessed / Reviewed by :	BJCHI Inspector	Client's Representative	Owner's Representative
PRINTED NAME			
SIGNATURE			
DATE			

F-QA-229 Rev.0 (09 Jul 2025)_Bolt Connection Gap Measurement 1 of 1